cBioPortal
FOR CANCER GENOMICS

# Tutorial 3
# Patient View

Investigate individual patients or samples in detail

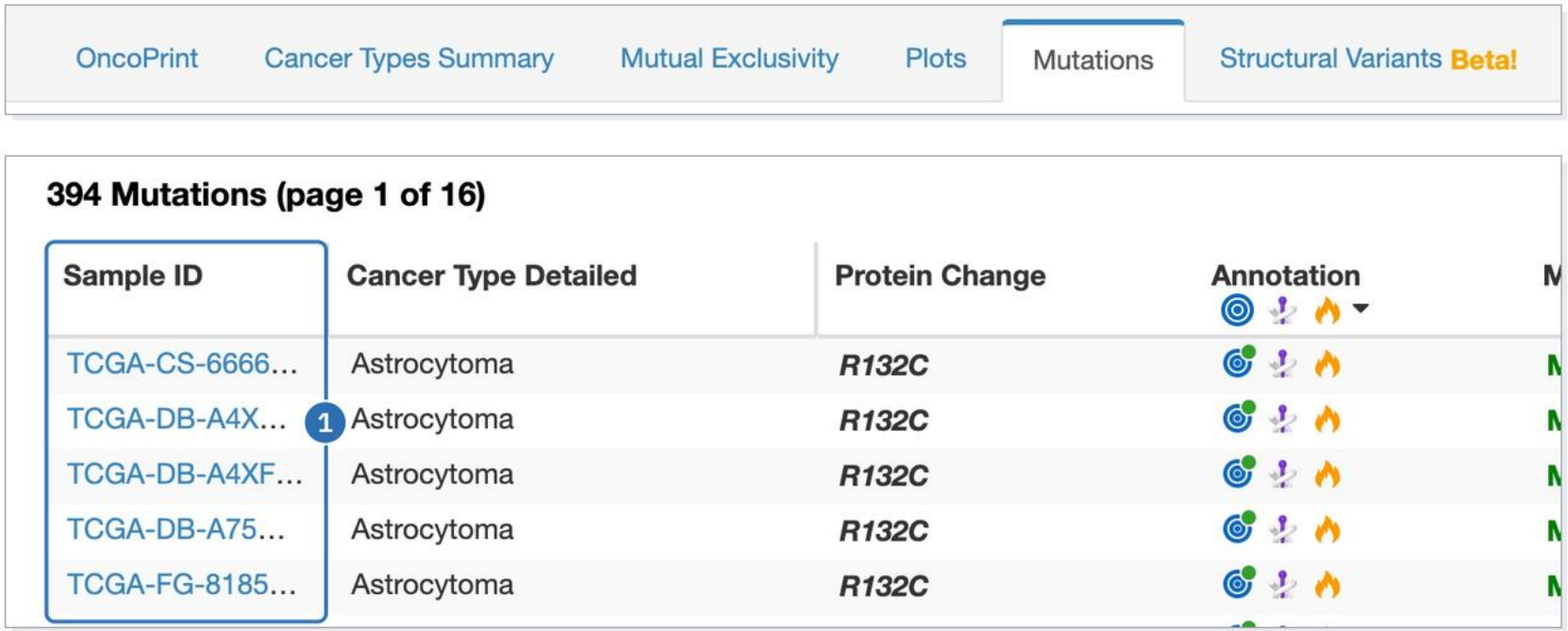
Last update: November 12, 2026

cbioportal.org/tutorials

# Tutorial objectives

- Show two ways to open the Patient View
- Walk through its tabs: Summary, Genomic Evolution, Pathways, Clinical Data, Pathology Report, Tissue Image and Slide Microscopy. Not every study has all of them: the tabs are study and patient specific, based on the data available
- Show how the data differ between studies: one sample per patient, or several samples over time
- Use the Patient View to answer a question about one tumor

# Opening the Patient View: click an ID




1. Anywhere you see a patient or sample ID, it links to the Patient View for that case. Here: the Mutations tab of the Tutorial 2 query. In the OncoPrint, hover over a sample and click its ID in the tooltip.

# Opening the Patient View: “View selected cases”




1. In the Study View, select the cases you want. Here: patients with 5 or 7 samples in Low-Grade Gliomas (UCSF, Science 2014), selected in the Number of Samples Per Patient chart.

2. 2 patients, 12 samples are selected.

3. Click “View selected cases” to open them in the Patient View, one patient at a time.

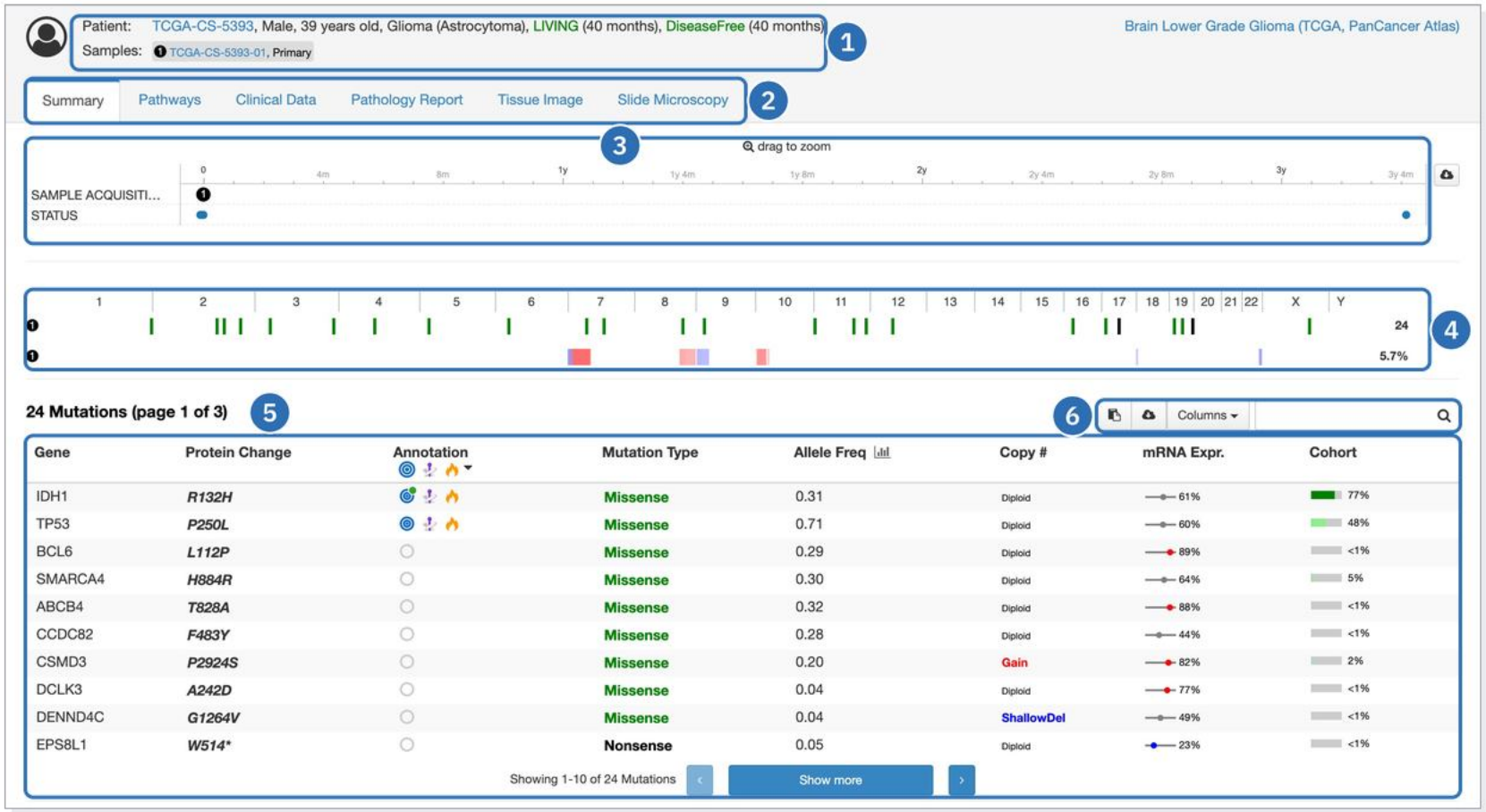


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# The Patient View

- However you get there, the Patient View opens on its Summary tab.
- The other tabs depend on the study: TCGA studies have pathology reports and tissue images; studies with several samples per patient have a Genomic Evolution tab.
- This tutorial uses three patients: one TCGA patient with one sample (Example 1), a patient with seven samples over time (Example 2), and a question about one endometrial tumor (Example 3).

# Example 1: Summary

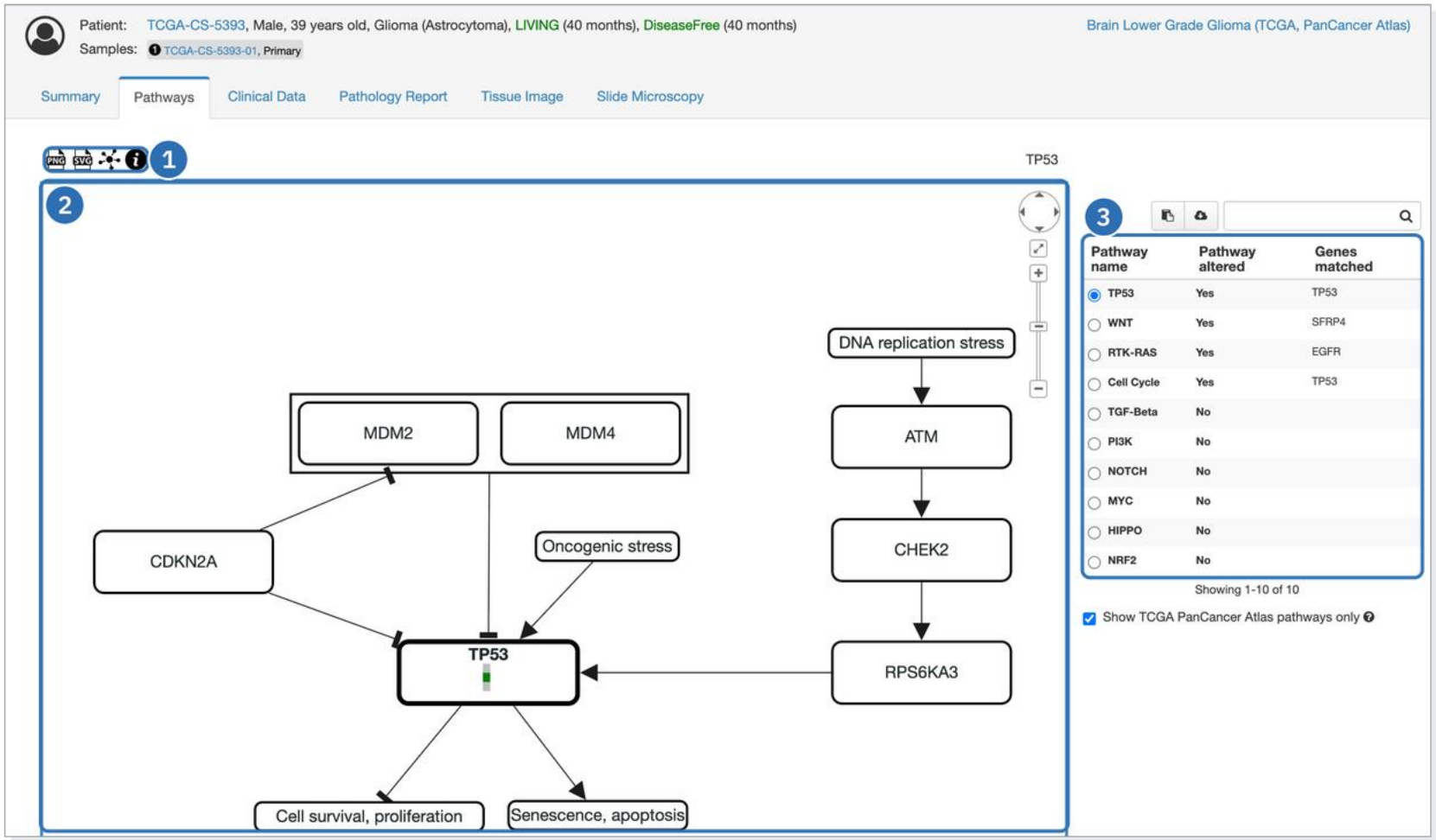



1. The patient and the sample. Hover over an ID for all its clinical data.
2. The tabs, which depend on the study.
3. A timeline of the patient's events, when the study has them. Drag to zoom.
4. Mutations (top row) and copy-number changes (bottom row) across the genome, with the mutation count and the fraction of the genome altered. Hover for details.
5. Every mutation, with its annotation, allele frequency, copy number, mRNA expression and how often the gene is mutated in the study (Cohort). Structural variants and copy-number alterations follow below.
6. Copy or download the table, choose columns, or search.



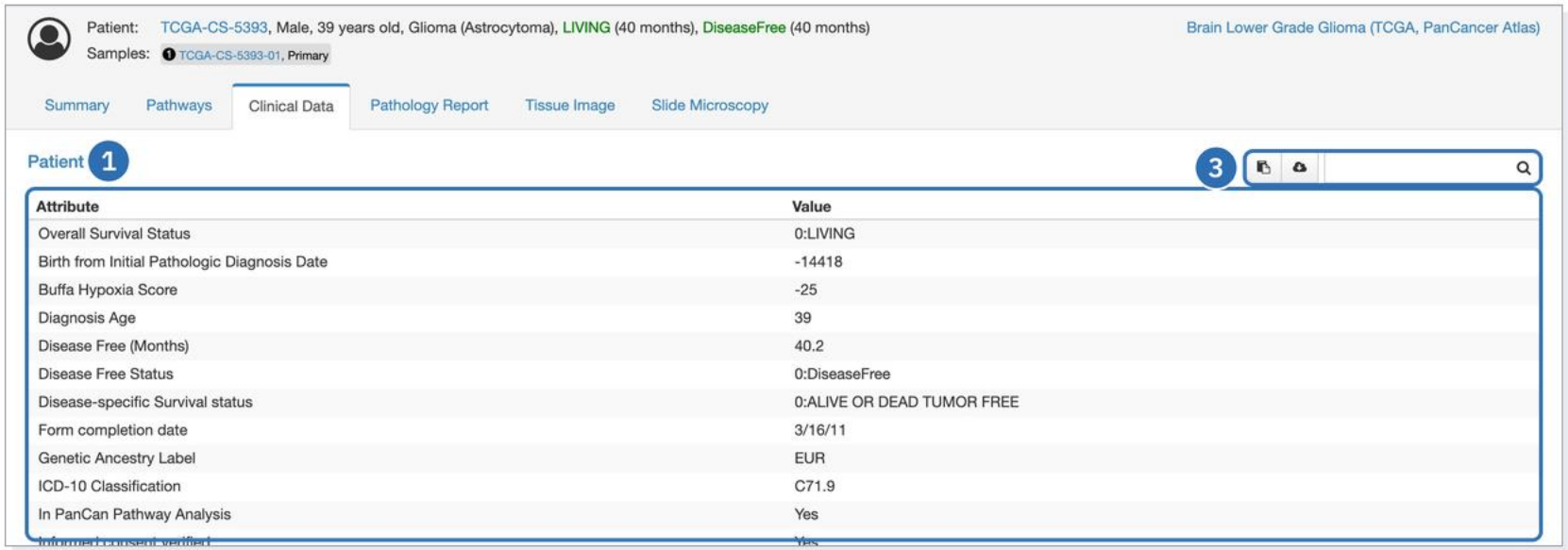
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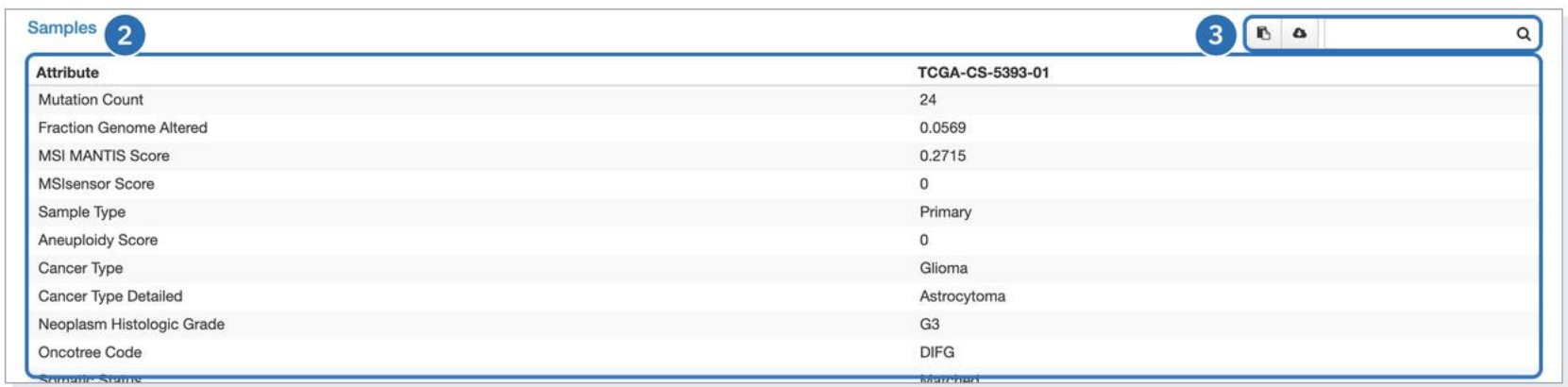
# Example 1: Pathways




1. Download the pathway (PNG, SVG) or open the editor; the i button explains the tools.
2. This patient's alterations in one of the TCGA pathways, from PathwayMapper. Here TP53, which is mutated.
3. Pathways ranked by whether this patient has alterations in them. Pick one to show it. See Tutorial 7.

# Example 1: Clinical Data



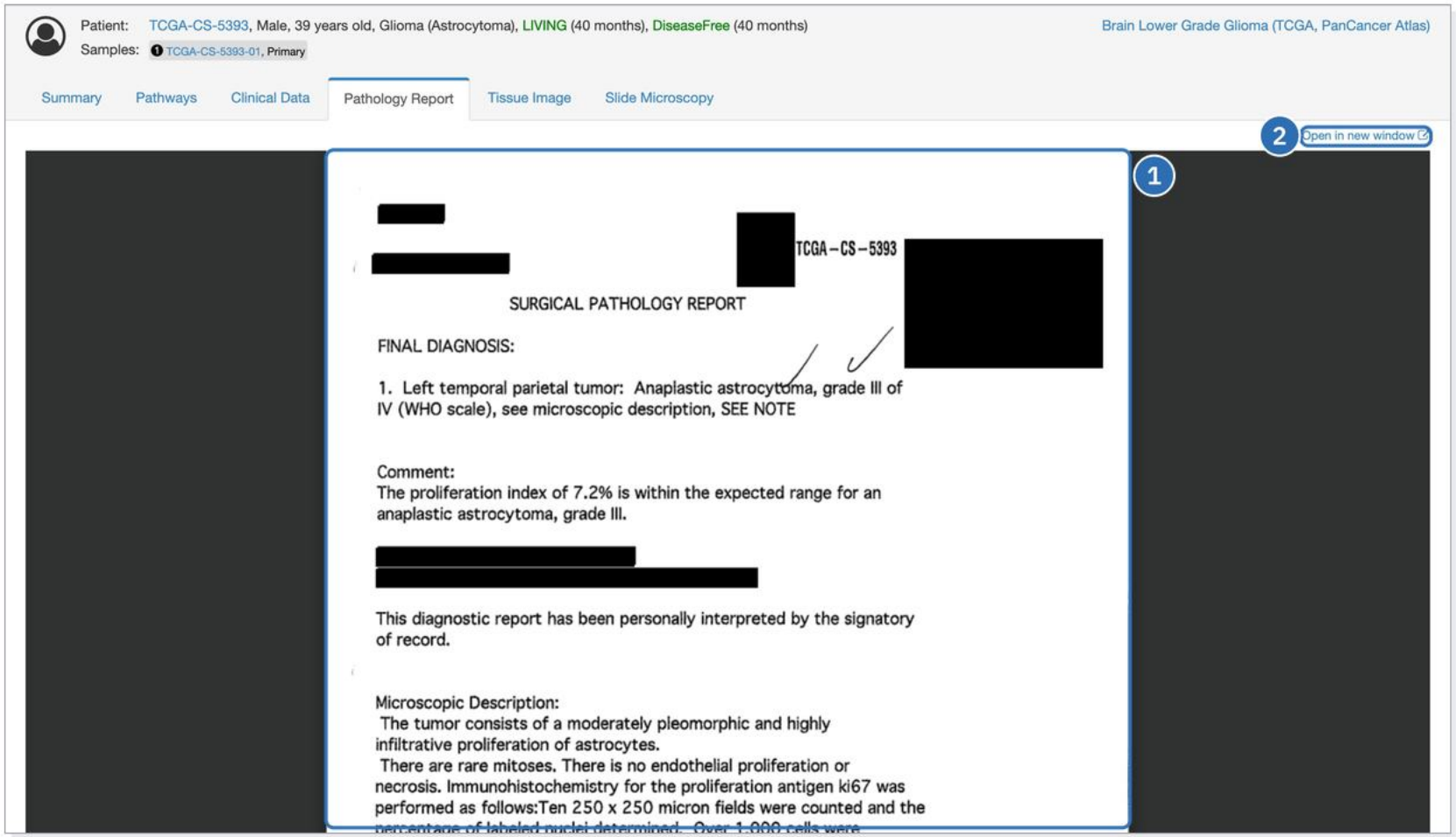




1. Every patient-level attribute in the study, e.g. survival status, diagnosis age, subtype.
2. Then the sample-level attributes: one column per sample.
3. Copy, download or search each table.



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# Example 1: Pathology Report

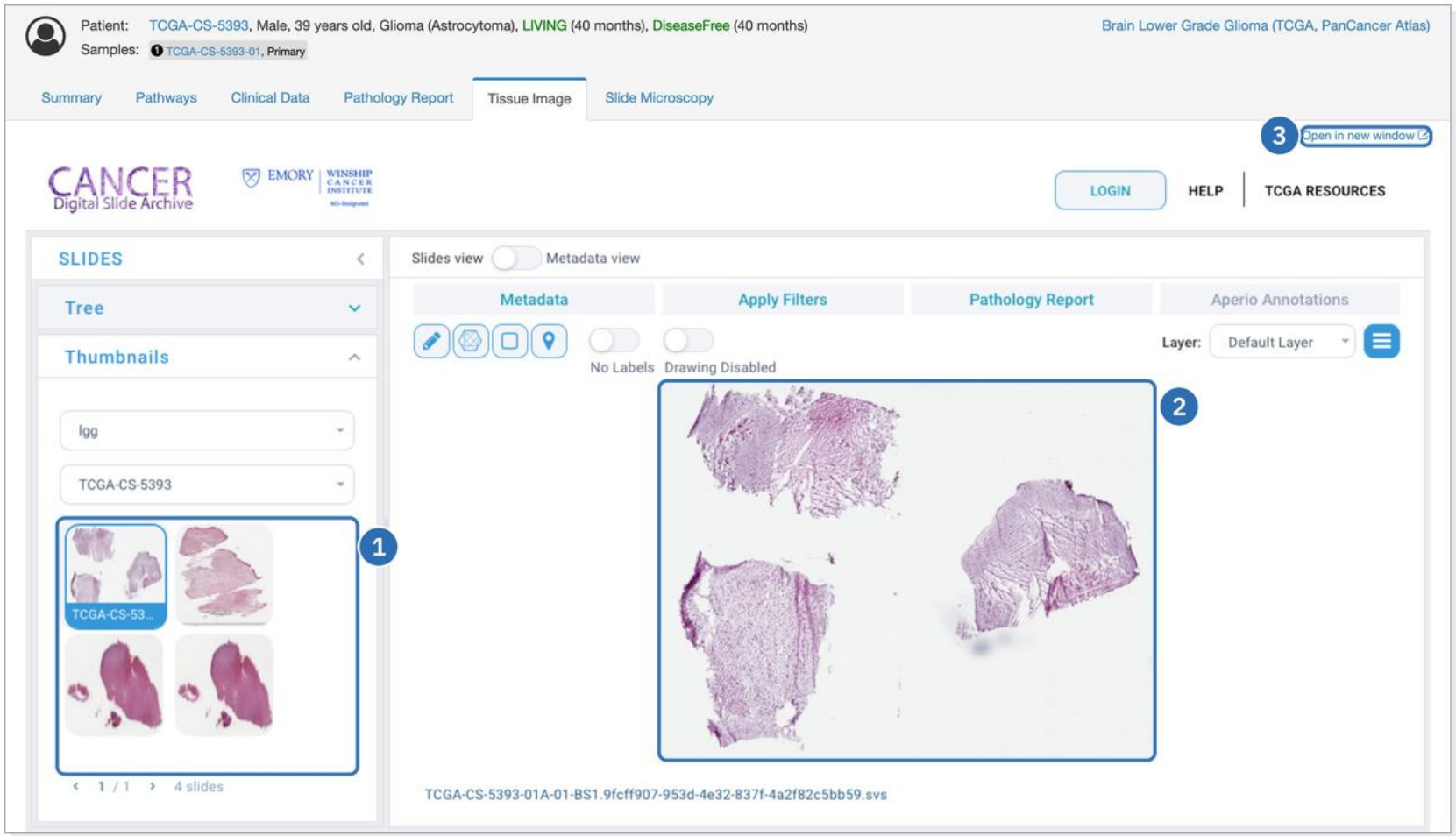



1. The original pathology report, de-identified. Only for TCGA studies.
2. Open it in a new window.



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# Example 1: Tissue Image

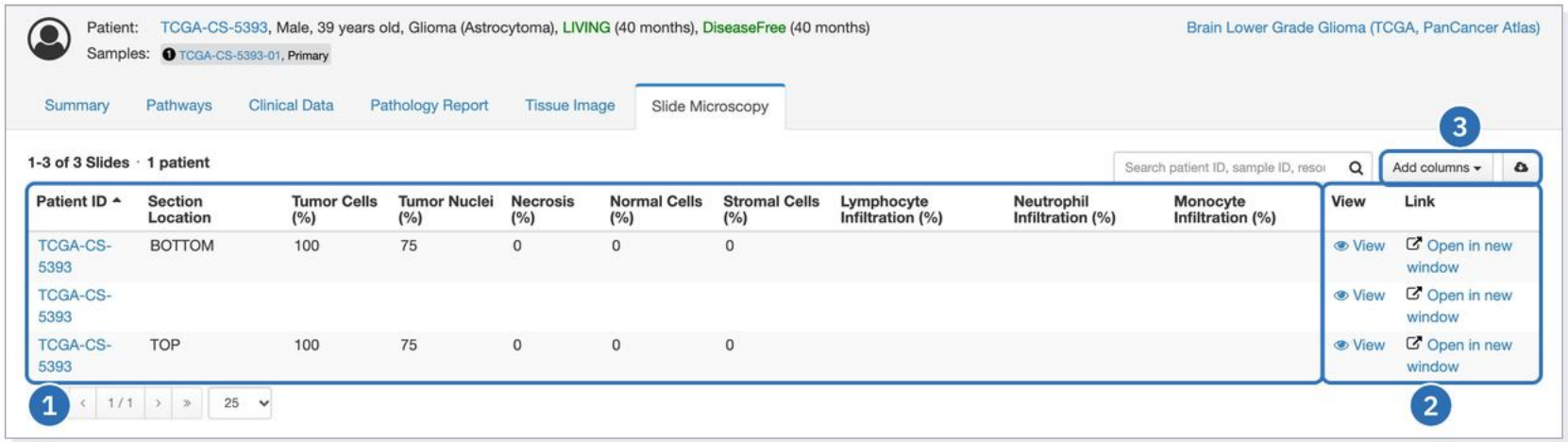



1. This patient's slides (4). Pick one; only for TCGA studies.
2. A zoomable image of the slide, from the Cancer Digital Slide Archive.
3. Open the viewer in a new window.



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# Example 1: Slide Microscopy




1. For studies with H&E slides in the NCI Imaging Data Commons: one row per slide, with pathology annotations (tumor cells, nuclei, necrosis).
2. View a slide here, or open it in the Imaging Data Commons viewer.
3. Add columns or download the table. Tutorial 1 shows the viewer.

# Example 2: Summary

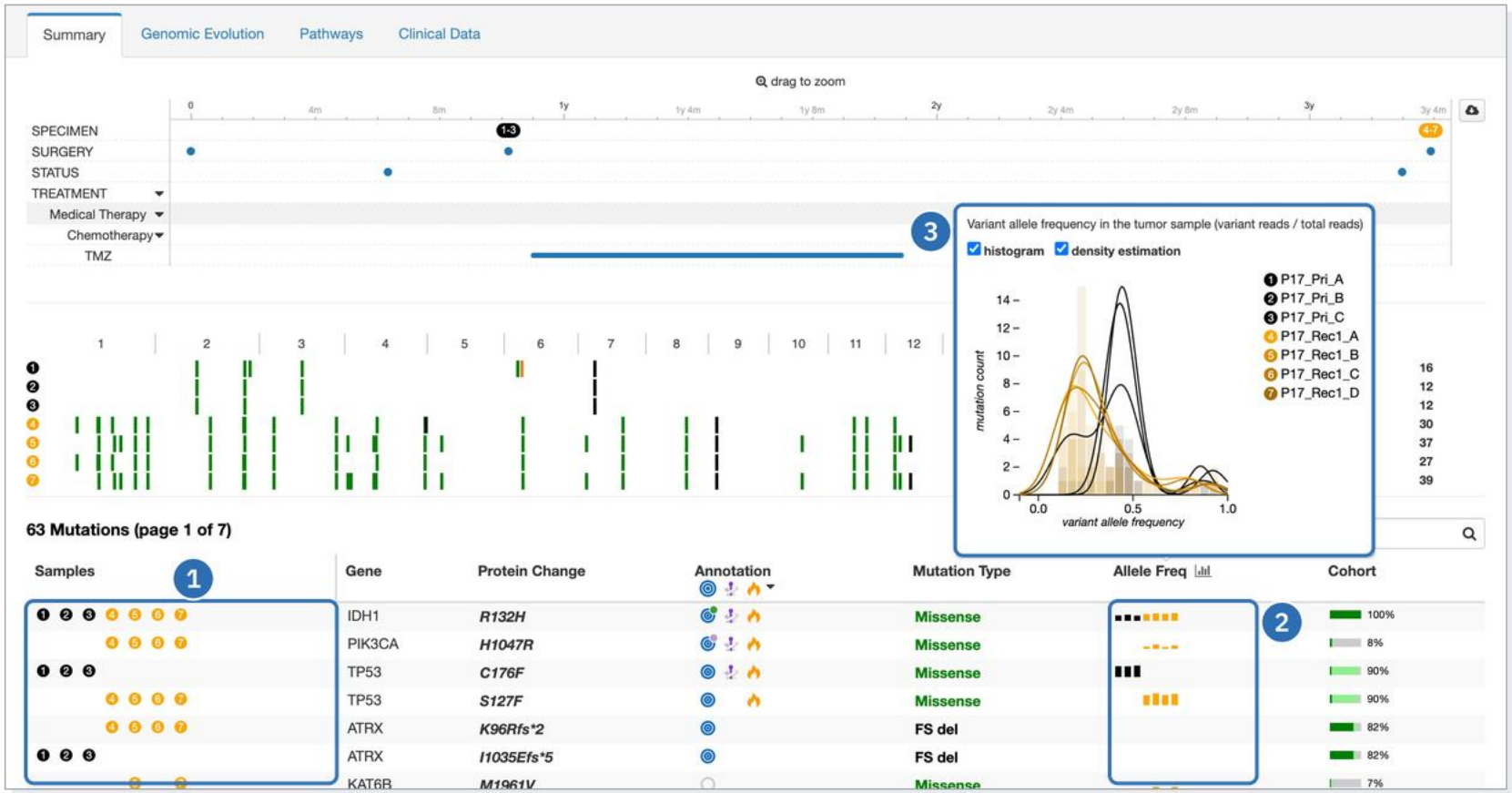



1. Patient P17 of Low-Grade Gliomas (UCSF, Science 2014) has 7 samples: 3 from the primary oligodendroglioma (black) and 4 from a recurrence as glioblastoma (orange). Hover over a sample for its data.
2. 1 of 2 patients: the arrows move to the other selected patient.
3. The timeline: surgeries, status (radiographic progression), samples and treatment (TMZ). Hover over an event for more details.
4. Mutations across the genome, one row per sample.
5. Every mutation. The Samples column shows which samples have it.



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# Example 2: allele frequencies

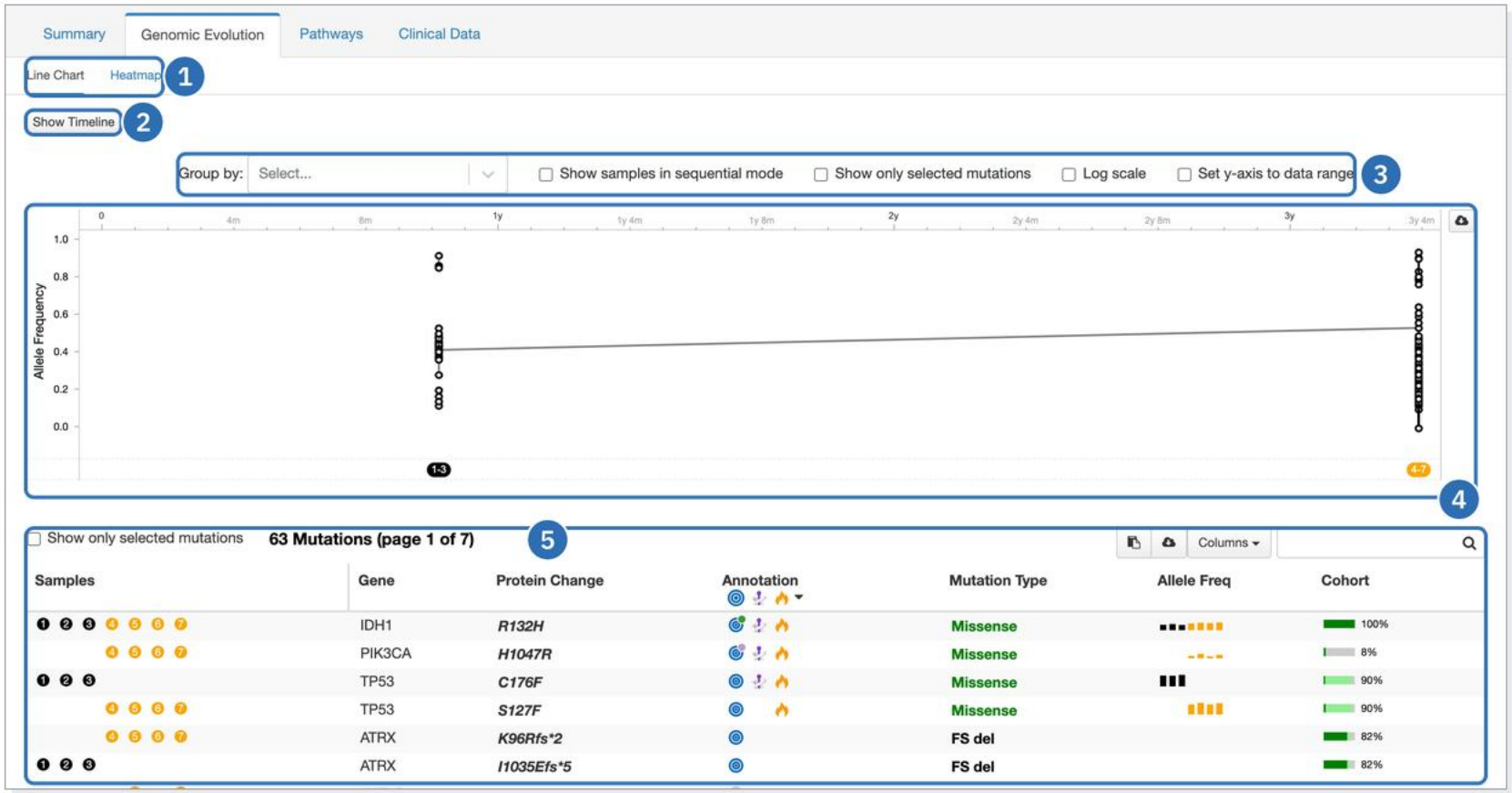



1. Which samples have each mutation.
2. Allele frequency: one bar per sample, its height the frequency.
3. Hover over the Allele Freq column header for the distribution of allele frequencies in each sample (histogram and density).



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# Example 2: Genomic Evolution, line chart

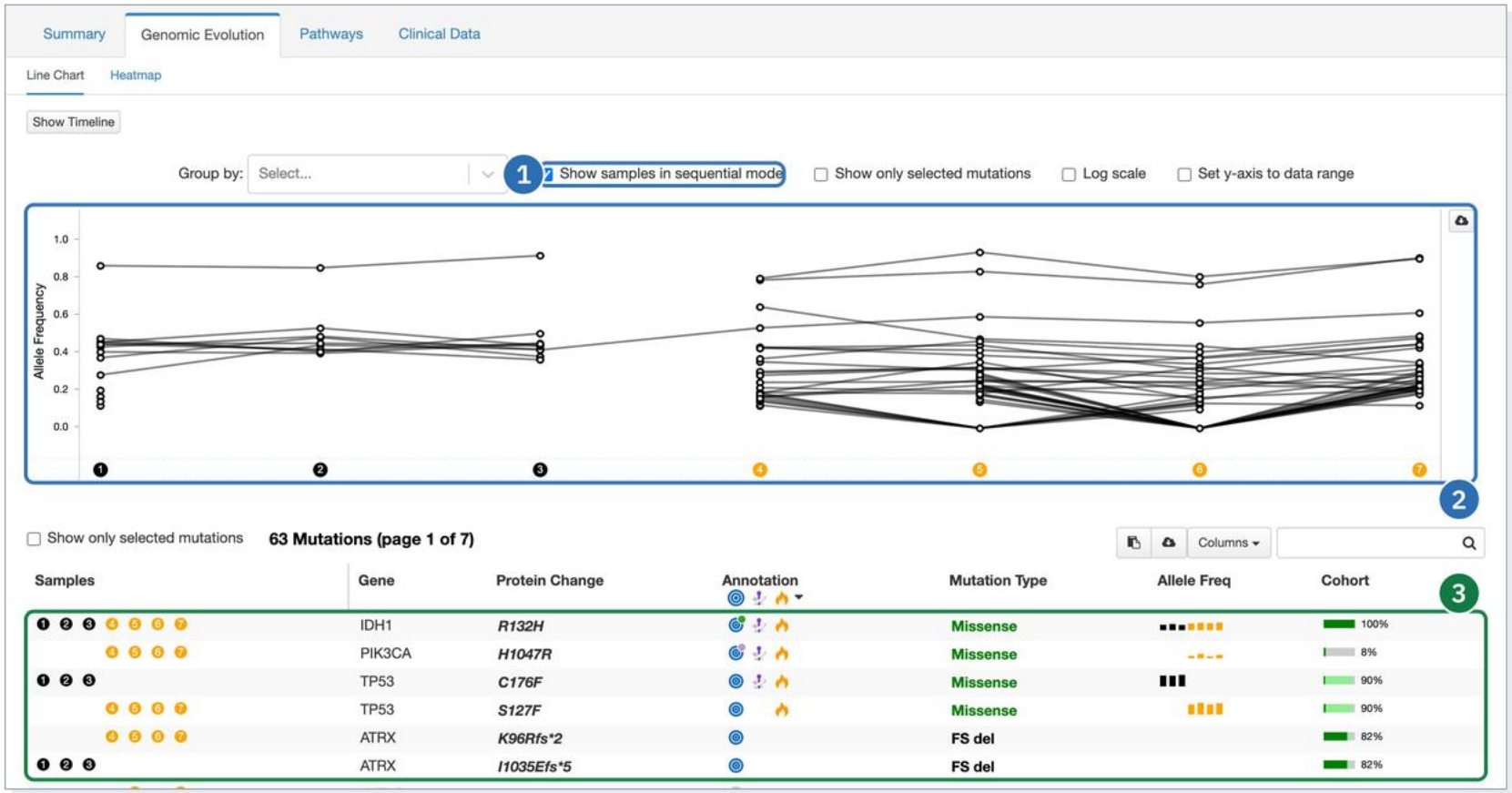



1. For patients with two or more samples: how allele frequencies change between samples. As a line chart or a heatmap.
2. “Show Timeline” shows the timeline above the chart.
3. Options to customize the chart.
4. Each dot is one mutation in one sample, placed in time; lines join the same mutation across samples.
5. The chart and table are linked: hover or click a mutation in one to highlight it in the other.



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# Example 2: Genomic Evolution, line chart




1. Show samples in sequential mode: the 7 samples evenly spaced, primary (1 to 3) then recurrence (4 to 7).

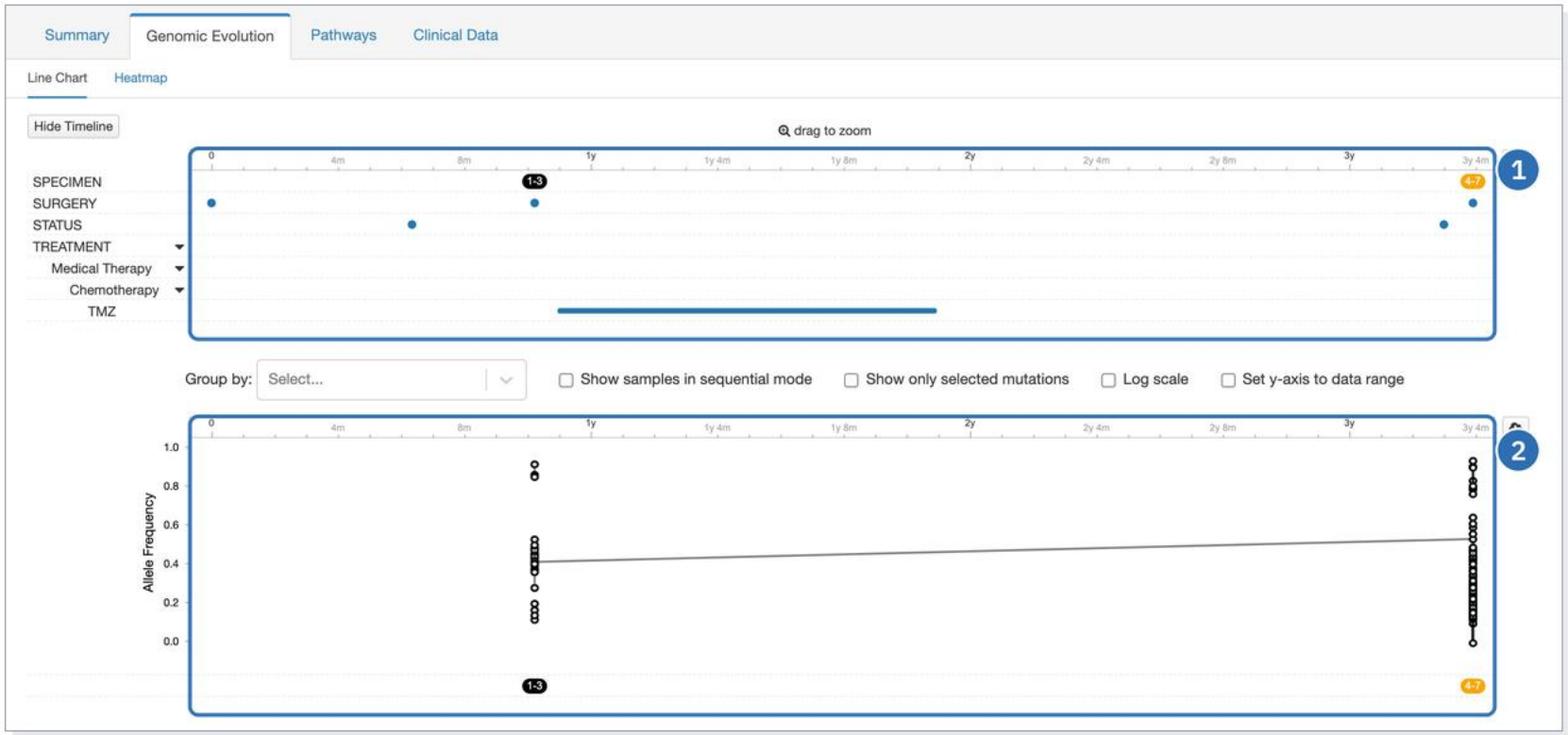
2. Most mutations are at similar frequencies within the primary and within the recurrence, respectively, but few are common to both.

3. Q: Did the recurrence grow from the cells of the primary tumor?
   A: Only IDH1 R132H is in all 7 samples. TP53 and ATRX are mutated in both, but by different mutations (TP53 C176F vs S127F); PIK3CA H1047R is only in the recurrence. The two branched off early.



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# Example 2: Genomic Evolution, line chart

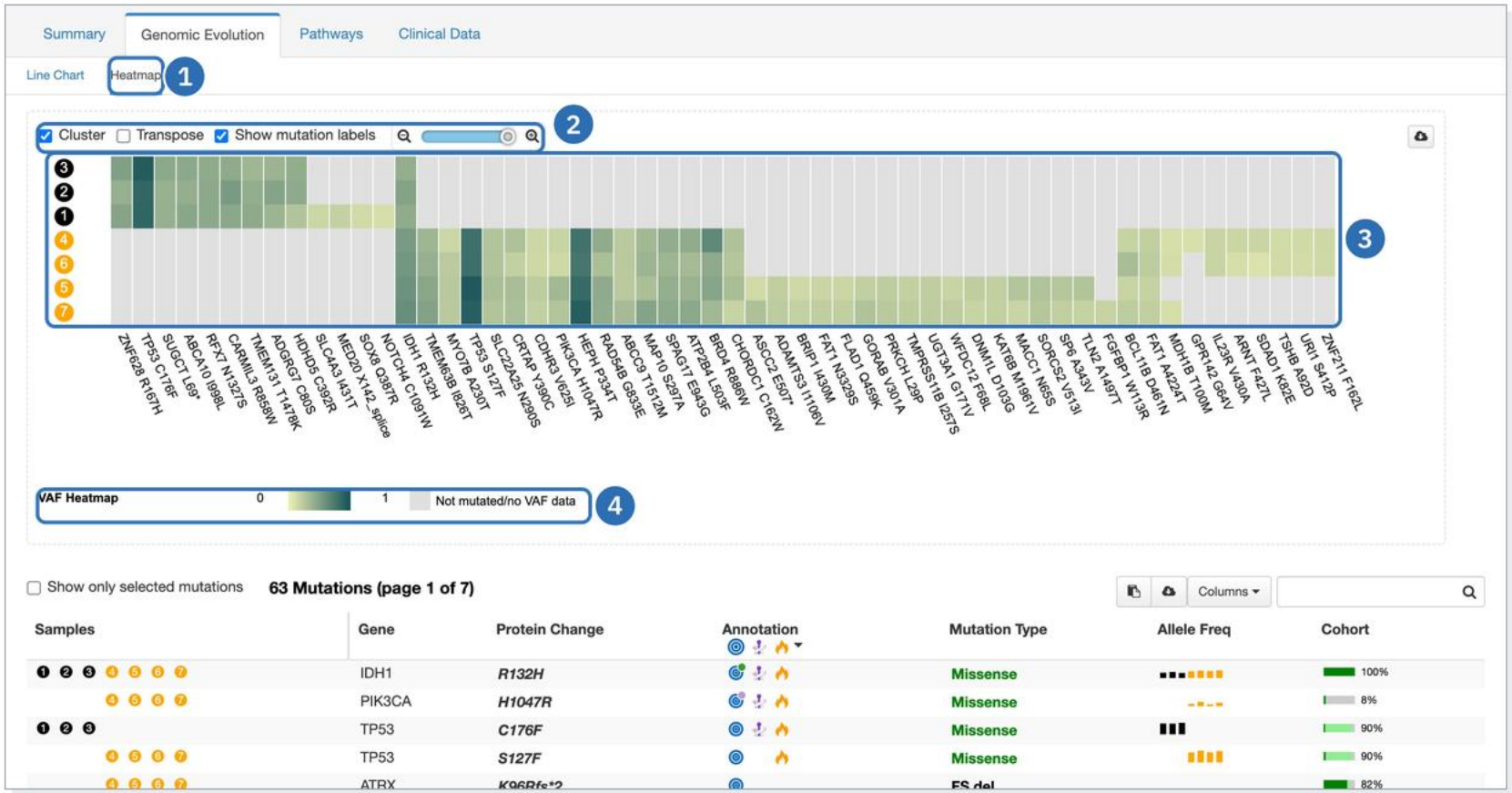



1. The patient's timeline, as on the Summary tab.
2. The allele frequencies on the same time axis: the recurrence samples were taken after TMZ treatment.



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# Example 2: Genomic Evolution, heatmap

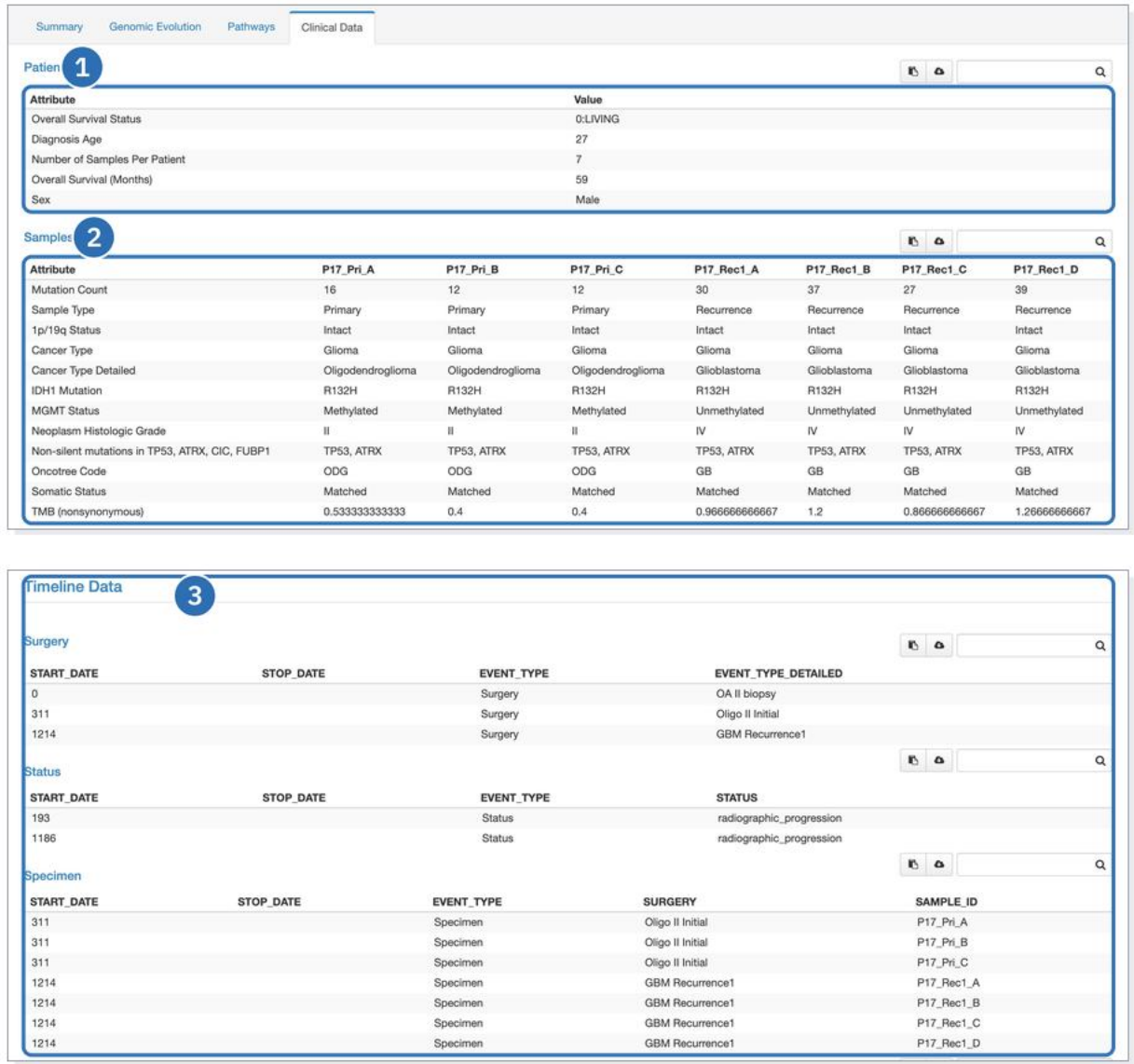



1. The Heatmap view.
2. Options to arrange and label the heatmap.
3. One row per sample, one column per mutation, colored by allele frequency. Clustering separates the primary samples (1 to 3) from the recurrence (4 to 7).
4. Allele frequency scale; grey is not mutated or no data.



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# Example 2: Clinical Data

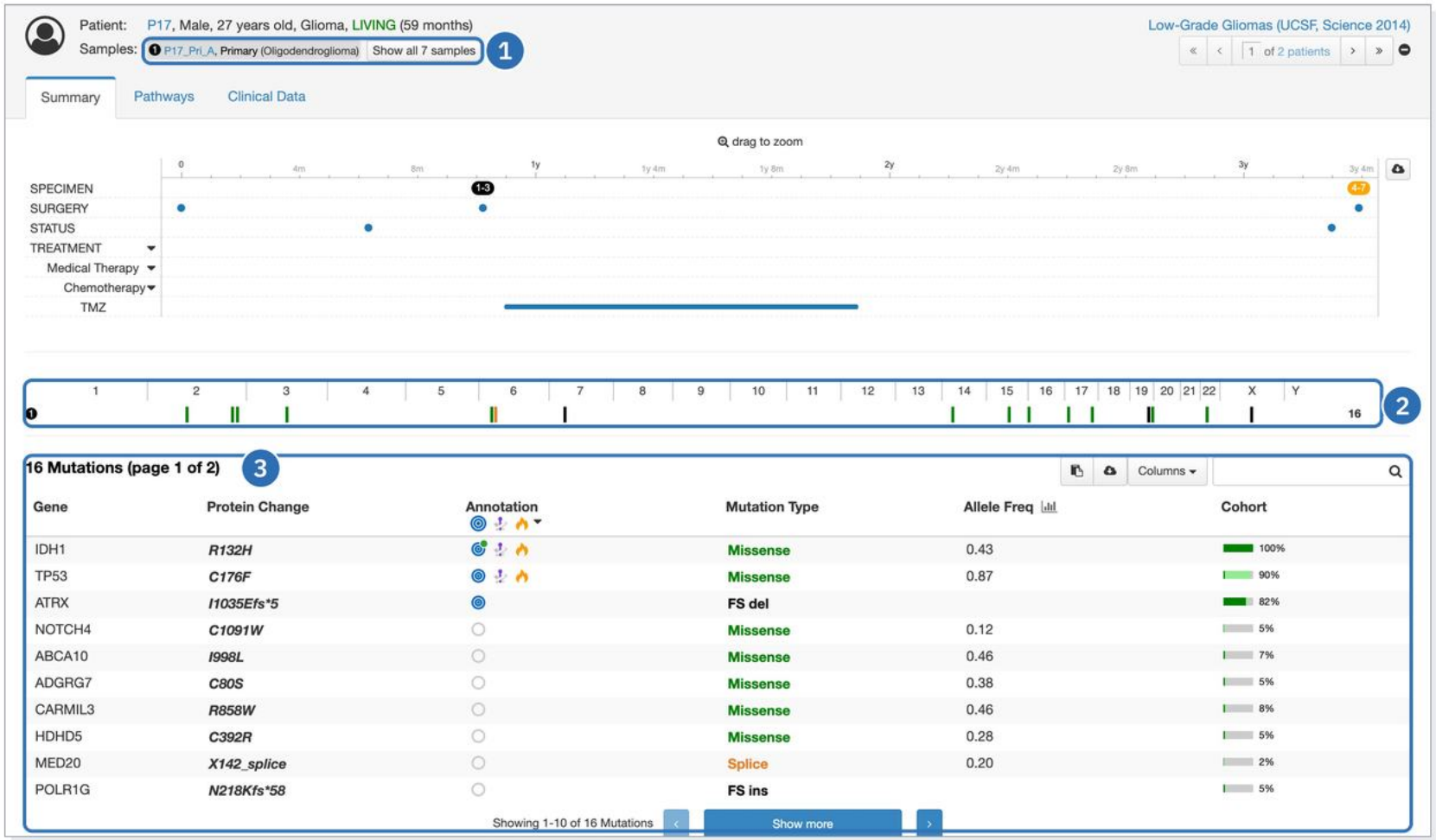



1. Patient-level attributes.
2. Sample-level attributes, one column per sample: type, histology, MGMT status, mutation count, TMB...
3. Timeline Data: the surgeries, status, specimens and treatments behind the timeline.



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# Example 2: one sample




1. Click a sample ID to open the view for that sample only (here P17_Pri_A). “Show all 7 samples” goes back.
2. That sample's mutations across the genome.
3. Its 16 mutations, with their allele frequency in this sample.

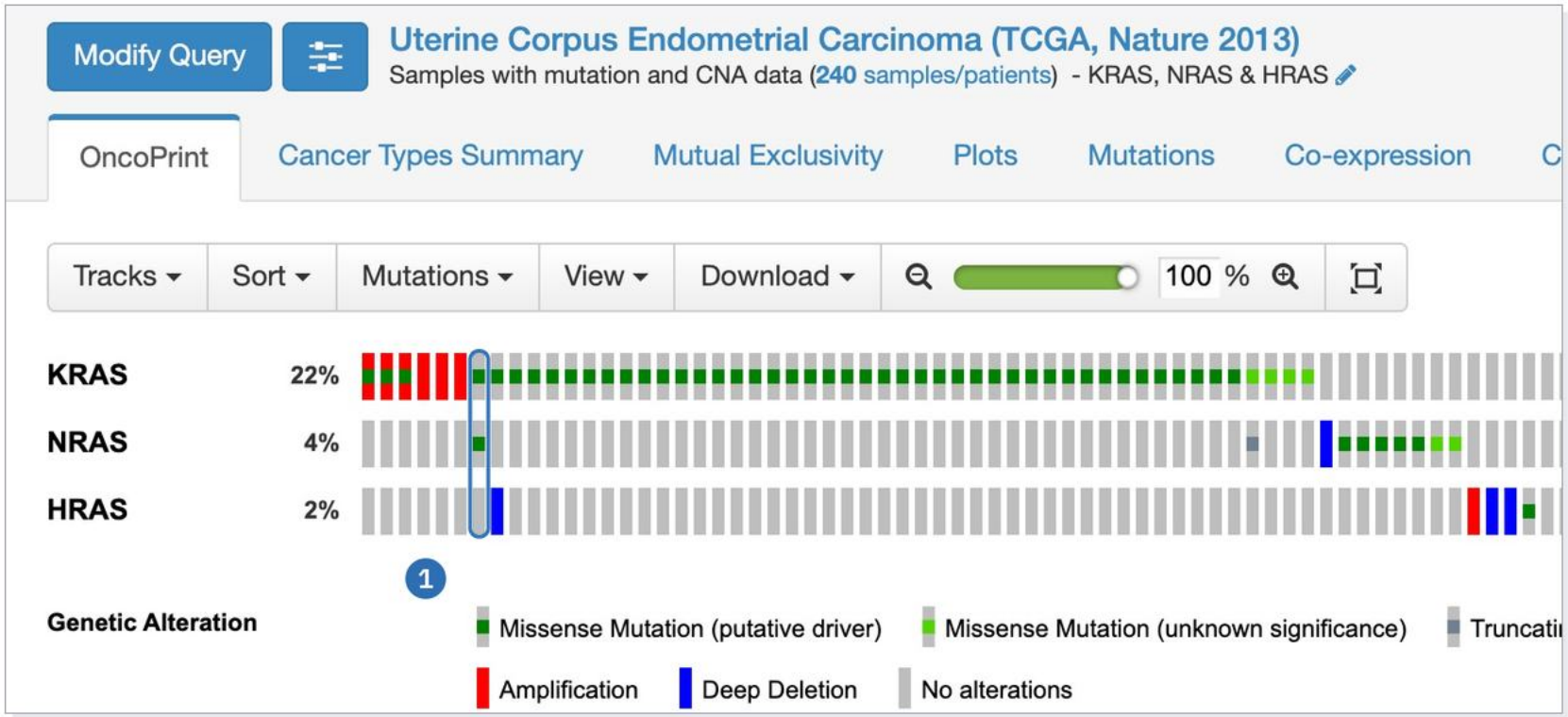


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# Example 3: RAS mutations in endometrial cancer

- Mutations in KRAS, NRAS and HRAS are usually mutually exclusive: a tumor rarely needs more than one.
- In Uterine Corpus Endometrial Carcinoma (TCGA, Nature 2013), query KRAS NRAS HRAS (see Tutorial 2).
- One sample has driver mutations in both KRAS and NRAS. The Patient View can tell us why.

# Example 3: OncoPrint

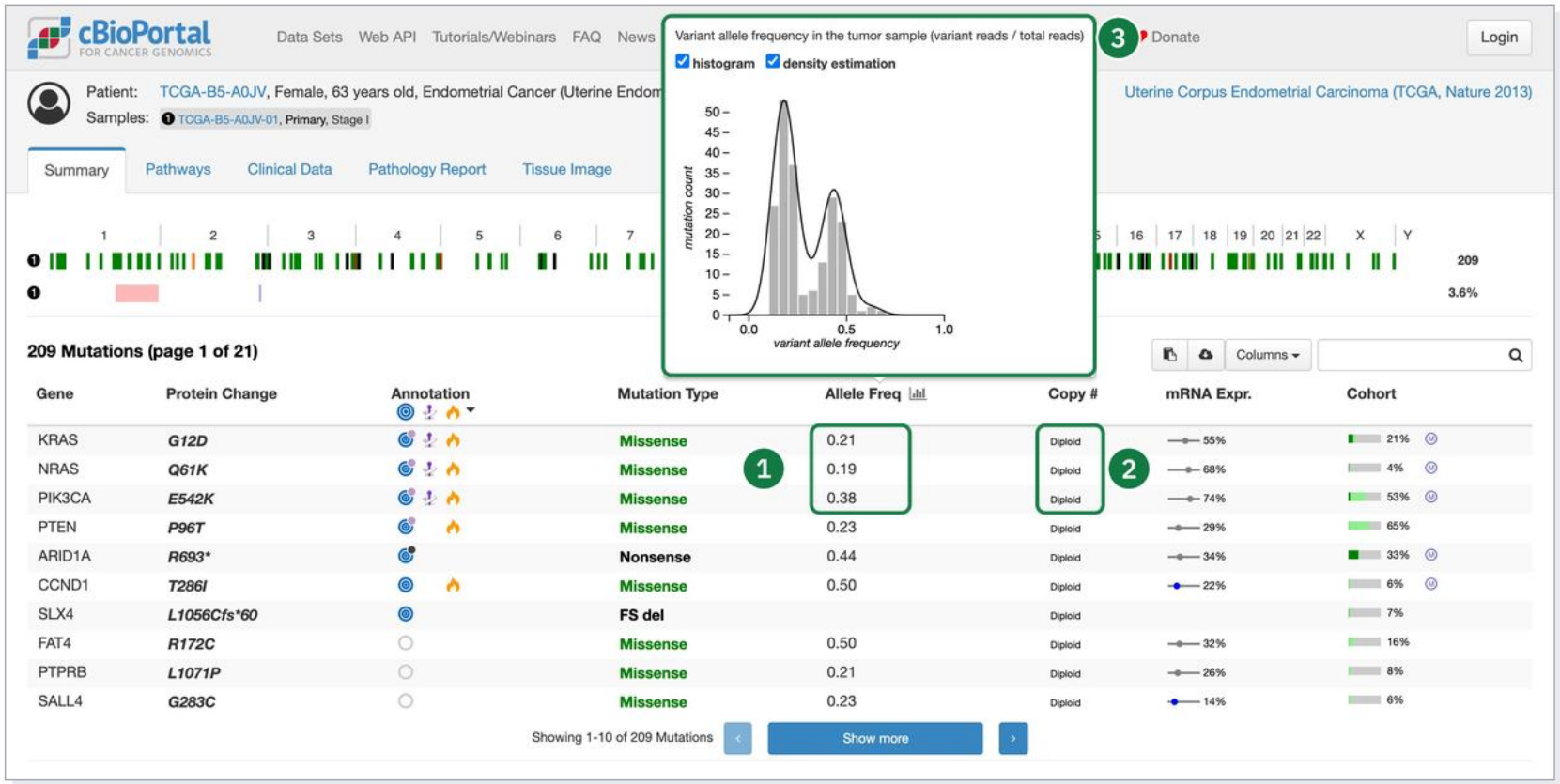



1. Alterations in the three genes are mostly mutually exclusive. One sample, TCGA-B5-A0JV, has driver mutations in both KRAS and NRAS (G12D and Q61K). Hover over its column and click the sample ID.



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# Example 3: Patient View




1. Q: How can one tumor have both KRAS and NRAS driver mutations?
   A: Look at the allele frequencies: KRAS G12D 0.21 and NRAS Q61K 0.19, but PIK3CA E542K 0.38, about twice as high.

2. A (continued): all three genes are diploid, so copy number does not explain the difference.

3. A (continued): the allele frequency histogram has two peaks (about 0.2 and 0.45): this suggests a clonal group of mutations, like PIK3CA, and a subclonal group, like KRAS and NRAS.



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# Example 3: what we learned

- PIK3CA E542K is probably in every tumor cell (clonal).
- KRAS G12D and NRAS Q61K are at about half that frequency: most likely each is in a different subclone of the same tumor, so no cell has both.
- The Patient View shows this from one sample's allele frequencies, copy number and their distribution.




# Questions?

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